

Brutal Practice Paper B07 - Reproduction in Plants, Comparing Quantities, Forests, Perimeter & Area

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A scientist sows 250 seeds and 80% of them germinate. Of the seedlings that germinate, $\frac{1}{5}$ die in the first week. How many seedlings survive the first week? [\[Reproduction in Plants\]](#)

- (A) 160
- (B) 144
- (C) 210
- (D) 180
- (E) 40
- (F) 190
- (G) 200
- (H) 50

2. A shirt marked Rs 800 is sold at a 15% discount, and then 5% GST is added on the discounted price. What is the final amount paid? [\[Comparing Quantities\]](#)

- (A) Rs 714
- (B) Rs 840
- (C) Rs 644
- (D) Rs 680
- (E) Rs 1428
- (F) Rs 700
- (G) Rs 720

3. In a forest food chain only 10% of energy passes to the next level. If the grass holds 50000 J, how much energy reaches the deer, and then the tiger one level above the deer? [\[Forests\]](#)

- (A) deer 500 J, tiger 5000 J
- (B) deer 5000 J, tiger 500 J
- (C) deer 5000 J, tiger 5000 J
- (D) deer 500 J, tiger 50 J
- (E) deer 5000 J, tiger 50 J
- (F) deer 45000 J, tiger 40500 J
- (G) deer 25000 J, tiger 12500 J

4. A rectangle is 18 cm long and 12 cm wide. A 2 cm wide path runs INSIDE it along all four edges. Find the area of the path. [\[Perimeter & Area\]](#)

- (A) 216 cm²
- (B) 128 cm²
- (C) 120 cm²
- (D) 112 cm²
- (E) 104 cm²
- (F) 208 cm²

5. A maize cob has 240 grains. If $\frac{5}{8}$ of the grains are viable seeds, how many grains will NOT grow into new plants? [\[Reproduction in Plants\]](#)
- (A) 90
 - (B) 96
 - (C) 120
 - (D) 192
 - (E) 135
 - (F) 150
6. A man buys 12 pens for Rs 144 and sells them at Rs 15 each. Find his profit percent. [\[Comparing Quantities\]](#)
- (A) 3%
 - (B) 50%
 - (C) 25%
 - (D) 15%
 - (E) 8%
 - (F) 30%
7. A forest patch of 400 trees has 35% tall canopy trees, 40% shorter understorey trees, and the rest are shrubs. How many shrubs are there? [\[Forests\]](#)
- (A) 160
 - (B) 90
 - (C) 140
 - (D) 120
 - (E) 75
 - (F) 100
8. A square has a perimeter of 64 cm. Find its area. [\[Perimeter & Area\]](#)
- (A) 4096 cm²
 - (B) 16 cm²
 - (C) 128 cm²
 - (D) 1024 cm²
 - (E) 512 cm²
 - (F) 256 cm²
9. In a wind-pollinated grass, each flower makes about 600 pollen grains and one spike carries 45 flowers. How many pollen grains does one spike make in all? [\[Reproduction in Plants\]](#)
- (A) 24000
 - (B) 645
 - (C) 27000
 - (D) 2700
 - (E) 26400
 - (F) 24300
 - (G) 270000

- 10.** In a class of 40 the ratio of boys to girls is 3:2. If 5 more girls join, what is the new ratio of boys to girls in lowest terms? [\[Comparing Quantities\]](#)
- (A) 8:7
 - (B) 6:5
 - (C) 3:2
 - (D) 2:1
 - (E) 8:9
 - (F) 7:8
 - (G) 4:3
- 11.** A fallen log of mass 24 kg loses $\frac{1}{8}$ of its mass to decomposition in one year. How much remains after the year, and how much was decomposed? [\[Forests\]](#)
- (A) 23 kg left, 1 kg decomposed
 - (B) 12 kg left, 12 kg decomposed
 - (C) 18 kg left, 6 kg decomposed
 - (D) 8 kg left, 16 kg decomposed
 - (E) 21 kg left, 3 kg decomposed
 - (F) 21 kg left, 21 kg decomposed
 - (G) 3 kg left, 21 kg decomposed
- 12.** The area of a rectangle is 96 cm² and its length is 12 cm. Find its perimeter. [\[Perimeter & Area\]](#)
- (A) 84 cm
 - (B) 20 cm
 - (C) 36 cm
 - (D) 40 cm
 - (E) 44 cm
 - (F) 200 cm
 - (G) 96 cm
- 13.** Every germinated bean seed shows 2 cotyledons. In a tray of 144 germinated bean seeds, the cotyledons are counted in dozens. How many dozen cotyledons are there altogether? [\[Reproduction in Plants\]](#)
- (A) 288
 - (B) 36
 - (C) 6
 - (D) 12
 - (E) 24
 - (F) 48
 - (G) 72
- 14.** A town's population rises 10% in the first year and then falls 10% in the second year. If it started at 50000, what is it after the two years? [\[Comparing Quantities\]](#)
- (A) 50500
 - (B) 49500
 - (C) 49000
 - (D) 55000
 - (E) 44550
 - (F) 45000
 - (G) 50000

- 15.** One tree releases about 120 kg of oxygen a year. A forest has 250 trees. How much oxygen does it release per year, and is that enough for 60 people who each need 500 kg of oxygen a year? [\[Forests\]](#)
- (A) 12000 kg, not enough
 - (B) 30000 kg, enough for 600 people
 - (C) 3000 kg, not enough
 - (D) 370 kg, not enough
 - (E) 30000 kg, not enough
 - (F) 30000 kg, more than enough
 - (G) 30000 kg, exactly enough
- 16.** A triangle has a base of 14 cm and a corresponding height of 9 cm. Find its area. [\[Perimeter & Area\]](#)
- (A) 126 cm²
 - (B) 31.5 cm²
 - (C) 63 cm²
 - (D) 189 cm²
 - (E) 49 cm²
 - (F) 112 cm²
 - (G) 23 cm²
- 17.** In an orchard of 320 flowers, 65% get pollinated, and of those pollinated only $\frac{3}{4}$ set fruit. How many flowers set fruit? [\[Reproduction in Plants\]](#)
- (A) 156
 - (B) 312
 - (C) 112
 - (D) 208
 - (E) 240
 - (F) 52
 - (G) 160
- 18.** A jacket costs Rs 2000. Its price is first increased by 25% and then decreased by 20%. What is the final price? [\[Comparing Quantities\]](#)
- (A) Rs 2250
 - (B) Rs 2100
 - (C) Rs 1800
 - (D) Rs 2000
 - (E) Rs 1900
 - (F) Rs 1600
- 19.** A forest of 1500 hectares loses 6% of its area each year to clearing. How many hectares remain after one year? [\[Forests\]](#)
- (A) 1590 hectares
 - (B) 1440 hectares
 - (C) 1410 hectares
 - (D) 1494 hectares
 - (E) 90 hectares
 - (F) 1269 hectares

20. A parallelogram and a triangle share the same base of 15 cm and the same height of 8 cm. By how much is the parallelogram's area greater than the triangle's? [\[Perimeter & Area\]](#)

- (A) 60 cm²
- (B) 240 cm²
- (C) 54 cm²
- (D) 90 cm²
- (E) 120 cm²
- (F) 30 cm²
- (G) 180 cm²

21. A dandelion scatters parachute-type seeds by wind. In a test, 18 of 24 released seeds travel more than 10 m. What percentage of the seeds is this? [\[Reproduction in Plants\]](#)

- (A) 18%
- (B) 150%
- (C) 72%
- (D) 80%
- (E) 25%
- (F) 75%
- (G) 60%

22. Find the simple interest on Rs 6000 at 8% per annum for 2 years, and the total amount returned.

[\[Comparing Quantities\]](#)

- (A) SI Rs 480, amount Rs 6960
- (B) SI Rs 960, amount Rs 6960
- (C) SI Rs 1920, amount Rs 7920
- (D) SI Rs 960, amount Rs 6000
- (E) SI Rs 96, amount Rs 6096
- (F) SI Rs 480, amount Rs 6480
- (G) SI Rs 960, amount Rs 7920

23. On a forest floor, 8 trees each drop 15 kg of leaves, and earthworms turn $\frac{2}{5}$ of all this litter into humus. How much humus forms? [\[Forests\]](#)

- (A) 60 kg
- (B) 96 kg
- (C) 48 kg
- (D) 40 kg
- (E) 30 kg
- (F) 72 kg
- (G) 120 kg

24. The circumference of a circle is 44 cm. Taking $\pi = \frac{22}{7}$, find its radius. [\[Perimeter & Area\]](#)

- (A) 14 cm
- (B) 22 cm
- (C) 7 cm
- (D) 11 cm
- (E) 3.5 cm
- (F) 21 cm

- 25.** A fern frond has 40 sori, and each sorus holds 64 spore-cases. How many spore-cases are there on the whole frond? [\[Reproduction in Plants\]](#)
- (A) 104
 - (B) 2304
 - (C) 2560
 - (D) 2240
 - (E) 256
 - (F) 25600
 - (G) 2056
- 26.** A number increased to 150% of itself becomes 270. What was the original number? [\[Comparing Quantities\]](#)
- (A) 120
 - (B) 135
 - (C) 162
 - (D) 180
 - (E) 405
 - (F) 90
- 27.** One large tree gives off 200 L of water vapour a day by transpiration. How many litres do 45 such trees give off in one week (7 days)? [\[Forests\]](#)
- (A) 6300 L
 - (B) 9000 L
 - (C) 45000 L
 - (D) 90000 L
 - (E) 56700 L
 - (F) 63000 L
 - (G) 1400 L
- 28.** A circle has a radius of 7 cm. Taking $\pi = \frac{22}{7}$, find its area. [\[Perimeter & Area\]](#)
- (A) 44 cm²
 - (B) 616 cm²
 - (C) 49 cm²
 - (D) 77 cm²
 - (E) 308 cm²
 - (F) 154 cm²
- 29.** A potato can be cut into pieces, each bearing 2 'eyes' that grow into new plants. A potato has 10 eyes. From 24 such potatoes, how many 2-eye pieces can be cut? [\[Reproduction in Plants\]](#)
- (A) 120
 - (B) 12
 - (C) 100
 - (D) 48
 - (E) 108
 - (F) 60
 - (G) 240

30. In a school 60% of the students are boys. If there are 480 girls, how many students are there in all?

[Comparing Quantities]

- (A) 1920
- (B) 1000
- (C) 1152
- (D) 720
- (E) 800
- (F) 1200

31. A forest absorbs about 22 kg of carbon dioxide per tree each year. To offset a car that emits 4400 kg of carbon dioxide a year, how many trees are needed? [Forests]

- (A) 20 trees
- (B) 440 trees
- (C) 200 trees
- (D) 96800 trees
- (E) 100 trees
- (F) 180 trees
- (G) 220 trees

32. A wire is bent into a square of side 11 cm and then re-bent into a circle. Taking $\pi = 22/7$, find the radius of the circle. [Perimeter & Area]

- (A) 7 cm
- (B) 5.5 cm
- (C) 22 cm
- (D) 21 cm
- (E) 44 cm
- (F) 14 cm
- (G) 11 cm

33. In a budding yeast culture the number of cells doubles every hour. Starting with 50 cells, how many cells are there after 3 hours? [Reproduction in Plants]

- (A) 800
- (B) 350
- (C) 150
- (D) 300
- (E) 400
- (F) 360

34. A trader marks his goods 40% above cost and then allows a 25% discount on the marked price. Find his profit percent. [Comparing Quantities]

- (A) 50%
- (B) 10%
- (C) 35%
- (D) 15%
- (E) 25%
- (F) 5%

- 35.** A forest survey finds birds, insects and mammals in the ratio 3:8:1. If 60 mammals are counted, how many insects are there? [\[Forests\]](#)
- (A) 60
 - (B) 240
 - (C) 30
 - (D) 540
 - (E) 180
 - (F) 480
 - (G) 720
- 36.** A rectangular field is 80 m long and 60 m wide. Find the cost of fencing it once around at Rs 25 per metre. [\[Perimeter & Area\]](#)
- (A) Rs 5600
 - (B) Rs 7000
 - (C) Rs 280
 - (D) Rs 6300
 - (E) Rs 14000
 - (F) Rs 120000
- 37.** A plant has 36 flowers, and each flower has 8 stamens and 1 pistil. How many stamens and how many pistils are there in total? [\[Reproduction in Plants\]](#)
- (A) 36 stamens, 288 pistils
 - (B) 44 stamens, 36 pistils
 - (C) 288 stamens, 288 pistils
 - (D) 288 stamens, 1 pistil
 - (E) 324 stamens, 36 pistils
 - (F) 288 stamens, 36 pistils
- 38.** Riya scored 90 out of 120 and Sam scored 105 out of 150. Whose percentage is higher? [\[Comparing Quantities\]](#)
- (A) Riya (80%)
 - (B) Riya (70%)
 - (C) Sam (80%)
 - (D) Sam (75%)
 - (E) Both 72.5%
 - (F) Sam (70%)
 - (G) Riya (75%)
- 39.** A forest guard plants 350 saplings; 80% survive the first year, and 90% of those survivors live through the second year. How many saplings are alive after two years? [\[Forests\]](#)
- (A) 315
 - (B) 245
 - (C) 70
 - (D) 504
 - (E) 238
 - (F) 252
 - (G) 280

- 40.** A square lawn of side 20 m has a 1 m wide path running all around it OUTSIDE. Find the area of the path. [\[Perimeter & Area\]](#)
- (A) 84 m²
 - (B) 44 m²
 - (C) 80 m²
 - (D) 400 m²
 - (E) 100 m²
 - (F) 168 m²
 - (G) 484 m²
- 41.** In a hand-pollination trial, 3 out of every 5 pollinated flowers produce seeds. If a researcher pollinates 85 flowers, how many produce seeds? [\[Reproduction in Plants\]](#)
- (A) 60
 - (B) 51
 - (C) 42
 - (D) 102
 - (E) 34
 - (F) 17
 - (G) 45
- 42.** The cost prices of two articles are in the ratio 5:7 and together they cost Rs 360. Find the cost of the cheaper article. [\[Comparing Quantities\]](#)
- (A) Rs 120
 - (B) Rs 150
 - (C) Rs 250
 - (D) Rs 135
 - (E) Rs 180
 - (F) Rs 210
- 43.** A forest fire spreads at 3 km per hour. How far does it spread in 40 minutes? (give the answer in km) [\[Forests\]](#)
- (A) 3 km
 - (B) 2 km
 - (C) 1.5 km
 - (D) 120 km
 - (E) 0.75 km
 - (F) 4 km
 - (G) 2.5 km
- 44.** A floor is 6 m long and 4 m wide. It is to be covered with square tiles of side 20 cm. How many tiles are needed? [\[Perimeter & Area\]](#)
- (A) 540 tiles
 - (B) 60 tiles
 - (C) 120 tiles
 - (D) 500 tiles
 - (E) 600 tiles
 - (F) 1200 tiles

45. A guava fruit holds 250 seeds. A bird eats 12 guavas and disperses all their seeds, but only 4% of dispersed seeds germinate. How many new plants grow? [\[Reproduction in Plants\]](#)

- (A) 1200
- (B) 120
- (C) 108
- (D) 300
- (E) 60
- (F) 30
- (G) 240

46. A bag holds red and blue balls in the ratio 4:5. If there are 36 red balls, how many balls are there in total? [\[Comparing Quantities\]](#)

- (A) 81
- (B) 36
- (C) 63
- (D) 72
- (E) 162
- (F) 45
- (G) 90

47. The dense canopy lets only 5% of sunlight reach the forest floor. If 2400 units of light strike the canopy, how many units reach the floor, and how many are blocked? [\[Forests\]](#)

- (A) 2280 reach, 120 blocked
- (B) 240 reach, 2160 blocked
- (C) 120 reach, 120 blocked
- (D) 480 reach, 1920 blocked
- (E) 600 reach, 1800 blocked
- (F) 120 reach, 2280 blocked
- (G) 1200 reach, 1200 blocked

48. A rectangle and a square have the same perimeter of 48 cm. The rectangle is 14 cm long. Which shape has the larger area, and by how much? [\[Perimeter & Area\]](#)

- (A) Square, by 4 cm²
- (B) Square, by 8 cm²
- (C) Equal areas
- (D) Square, by 14 cm²
- (E) Rectangle, by 4 cm²
- (F) Rectangle, by 6 cm²
- (G) Square, by 2 cm²

49. A coconut is dispersed by water. In a lagoon, 7 of 20 floating coconuts reach a new island. Express this success as a fraction in lowest terms and as a percentage. [\[Reproduction in Plants\]](#)

- (A) $\frac{7}{20} = 70\%$
- (B) $\frac{7}{20} = 3.5\%$
- (C) $\frac{7}{20} = 35\%$
- (D) $\frac{7}{13} = 54\%$
- (E) $\frac{1}{3} = 33\%$
- (F) $\frac{13}{20} = 65\%$
- (G) $\frac{7}{20} = 7\%$

50. A cycle bought for Rs 3200 is sold at a loss of 12.5%. Find the selling price. [\[Comparing Quantities\]](#)

- (A) Rs 3000
- (B) Rs 400
- (C) Rs 2880
- (D) Rs 3600
- (E) Rs 2520
- (F) Rs 2800
- (G) Rs 2400

51. A herd of 24 deer each eat 6 kg of leaves per day. How many kilograms of leaves does the whole herd eat during the month of April (30 days)? [\[Forests\]](#)

- (A) 2160 kg
- (B) 180 kg
- (C) 4320 kg
- (D) 43200 kg
- (E) 720 kg
- (F) 144 kg
- (G) 3888 kg

52. An L-shaped park is made by taking a 30 m by 20 m rectangle and removing a 10 m by 8 m rectangle from one corner. Find the remaining area. [\[Perimeter & Area\]](#)

- (A) 520 m²
- (B) 680 m²
- (C) 468 m²
- (D) 600 m²
- (E) 440 m²
- (F) 540 m²
- (G) 80 m²

53. In a garden, $\frac{5}{8}$ of the 96 flowers are bisexual and the rest are unisexual. How many unisexual flowers are there? [\[Reproduction in Plants\]](#)

- (A) 72
- (B) 48
- (C) 40
- (D) 60
- (E) 30
- (F) 36
- (G) 12

54. 35% of a number is 70. What is 80% of the same number? [\[Comparing Quantities\]](#)

- (A) 160
- (B) 144
- (C) 56
- (D) 140
- (E) 175
- (F) 200

55. Without tree cover, soil erodes at 12 tonnes per hectare each year. A cleared patch is 25 hectares. How much soil is lost in 3 years? [\[Forests\]](#)

- (A) 9000 tonnes
- (B) 40 tonnes
- (C) 900 tonnes
- (D) 100 tonnes
- (E) 300 tonnes
- (F) 600 tonnes
- (G) 810 tonnes

56. Find the cost of fencing a circular flower bed of radius 14 m at Rs 10 per metre. Take $\pi = 22/7$. [\[Perimeter & Area\]](#)

- (A) Rs 6160
- (B) Rs 1760
- (C) Rs 880
- (D) Rs 792
- (E) Rs 440
- (F) Rs 88
- (G) Rs 1232

57. In an apple orchard the ratio of pollinated to unpollinated flowers is 7:3. If there are 210 unpollinated flowers, how many flowers are there in total? [\[Reproduction in Plants\]](#)

- (A) 630
- (B) 900
- (C) 490
- (D) 700
- (E) 100
- (F) 770

58. A shopkeeper sells an item for Rs 575 at a profit of 15%. Find the cost price. [\[Comparing Quantities\]](#)

- (A) Rs 550
- (B) Rs 460
- (C) Rs 450
- (D) Rs 525
- (E) Rs 500
- (F) Rs 488.75
- (G) Rs 489

59. A tree takes up 50 kg of nutrients a year and returns 60% of them to the soil through fallen leaves. How many kilograms does it keep in its wood each year? [\[Forests\]](#)

- (A) 50 kg
- (B) 25 kg
- (C) 30 kg
- (D) 20 kg
- (E) 10 kg
- (F) 18 kg

60. A triangle has an area of 84 cm² and a base of 24 cm. Find its height. [\[Perimeter & Area\]](#)

- (A) 6 cm
- (B) 3.5 cm
- (C) 14 cm
- (D) 168 cm
- (E) 21 cm
- (F) 7 cm

61. A sunflower head holds about 1000 seeds. Birds eat 12% of them, and half of the remaining seeds are sown. How many seeds are sown? [\[Reproduction in Plants\]](#)

- (A) 880
- (B) 440
- (C) 120
- (D) 396
- (E) 500
- (F) 560
- (G) 60

62. Two discounts of 20% and 10% are given one after the other on a price of Rs 1000. What single discount percent gives the same final price? [\[Comparing Quantities\]](#)

- (A) 32%
- (B) 56%
- (C) 28%
- (D) 72%
- (E) 70%
- (F) 30%

63. In a food web, one tiger needs the energy of 8 deer each month, and each deer needs the grass-energy of 100 kg of grass each month. How many kilograms of grass-energy support one tiger per month? [\[Forests\]](#)

- (A) 720 kg
- (B) 1000 kg
- (C) 8 kg
- (D) 80 kg
- (E) 108 kg
- (F) 800 kg

64. A room is 5 m long, 4 m wide and 3 m high. Its four walls (not the floor or ceiling) are to be painted. Find the area to be painted. [\[Perimeter & Area\]](#)

- (A) 81 m²
- (B) 20 m²
- (C) 108 m²
- (D) 60 m²
- (E) 54 m²
- (F) 94 m²

- 65.** A strawberry plant sends out 6 runners, and each runner grows 4 daughter plants. If a farmer has 15 such parent plants, how many daughter plants form in total? [\[Reproduction in Plants\]](#)
- (A) 375
 - (B) 360
 - (C) 144
 - (D) 240
 - (E) 150
 - (F) 324
- 66.** A 500 g alloy contains copper and zinc in the ratio 3:2. How many grams of zinc must be ADDED to make the ratio of copper to zinc 1:1? [\[Comparing Quantities\]](#)
- (A) 200 g
 - (B) 150 g
 - (C) 250 g
 - (D) 50 g
 - (E) 90 g
 - (F) 300 g
 - (G) 100 g
- 67.** A forest soaks up rain like a sponge. Of 500 mm of rainfall, 70% soaks into the ground and the rest runs off. How much soaks in, and how much runs off? [\[Forests\]](#)
- (A) 250 mm soak in, 250 mm run off
 - (B) 70 mm soak in, 30 mm run off
 - (C) 350 mm soak in, 150 mm run off
 - (D) 350 mm soak in, 350 mm run off
 - (E) 350 mm soak in, 150 mm soak in
 - (F) 300 mm soak in, 200 mm run off
- 68.** The length of a rectangle is twice its breadth, and its perimeter is 72 cm. Find its area. [\[Perimeter & Area\]](#)
- (A) 288 cm²
 - (B) 576 cm²
 - (C) 324 cm²
 - (D) 432 cm²
 - (E) 72 cm²
 - (F) 200 cm²
 - (G) 144 cm²
- 69.** Pollen of a certain flower stays viable for 2 days. A bee arrives 30 hours after the anther opened. For how many more hours will the pollen stay viable? (1 day = 24 hours) [\[Reproduction in Plants\]](#)
- (A) 12 hours
 - (B) 48 hours
 - (C) 18 hours
 - (D) 22 hours
 - (E) 78 hours
 - (F) 36 hours

70. The price of petrol rises from Rs 80 to Rs 100 per litre. By what percent did the price rise? [\[Comparing Quantities\]](#)

- (A) 25%
- (B) 50%
- (C) 20%
- (D) 100%
- (E) 16.6%
- (F) 40%
- (G) 80%

71. A country loses 2400 square km of forest every year. At this rate, how many years will it take to lose 16800 square km of forest? [\[Forests\]](#)

- (A) 14 years
- (B) 5 years
- (C) 70 years
- (D) 6 years
- (E) 40320000 years
- (F) 7 years
- (G) 8 years

72. A rectangular sheet is 24 cm by 16 cm. The largest possible square is cut from it. Find the area of the leftover piece. [\[Perimeter & Area\]](#)

- (A) 128 cm²
- (B) 64 cm²
- (C) 160 cm²
- (D) 384 cm²
- (E) 192 cm²
- (F) 144 cm²

73. A papaya field has male and female plants in the ratio 2:3, and only female plants bear fruit. If there are 150 plants in all, how many plants bear fruit? [\[Reproduction in Plants\]](#)

- (A) 30
- (B) 75
- (C) 90
- (D) 60
- (E) 81
- (F) 45
- (G) 120

74. A student scored 60% in term 1 and 75% in term 2, each out of the same total of 200 marks. By what percent did the actual marks increase from term 1 to term 2? [\[Comparing Quantities\]](#)

- (A) 18%
- (B) 25%
- (C) 15%
- (D) 20%
- (E) 30%
- (F) 12.5%
- (G) 50%

75. A forest's layers have heights in the ratio 8:6:3:1 for the emergent, canopy, understorey and floor layers. If the emergent layer is 40 m tall, how tall is the understorey layer? [\[Forests\]](#)

- (A) 30 m
- (B) 120 m
- (C) 5 m
- (D) 15 m
- (E) 24 m
- (F) 18 m

76. Two circles have radii 7 cm and 14 cm. Taking $\pi = \frac{22}{7}$, by how much is the larger circle's area greater than the smaller one's? [\[Perimeter & Area\]](#)

- (A) 231 cm²
- (B) 616 cm²
- (C) 308 cm²
- (D) 770 cm²
- (E) 154 cm²
- (F) 462 cm²
- (G) 924 cm²

77. In a germination test, Tray A has 45 of 60 seeds sprout and Tray B has 56 of 80 seeds sprout. Which tray has the higher germination percentage? [\[Reproduction in Plants\]](#)

- (A) Both equal (75%)
- (B) Both equal (72.5%)
- (C) Tray A (80%)
- (D) Tray B (75%)
- (E) Tray A (70%)
- (F) Tray A (75%)

78. A man spends 75% of his income and saves Rs 4500. What is his monthly income? [\[Comparing Quantities\]](#)

- (A) Rs 16200
- (B) Rs 3375
- (C) Rs 13500
- (D) Rs 22500
- (E) Rs 6000
- (F) Rs 18000

79. A protected forest had 800 elephants. The number fell by 15% due to habitat loss, then a conservation drive raised the new number by 20%. How many elephants are there now? [\[Forests\]](#)

- (A) 920
- (B) 680
- (C) 800
- (D) 816
- (E) 1632
- (F) 760
- (G) 840

- 80.** A rectangular garden is 50 m by 30 m. Two paths, each 2 m wide, cross through the middle - one along the length and one along the breadth. Find the total area of the paths. [\[Perimeter & Area\]](#)
- (A) 160 m²
 - (B) 100 m²
 - (C) 156 m²
 - (D) 164 m²
 - (E) 200 m²
 - (F) 312 m²
 - (G) 152 m²
- 81.** A maize cob has kernels in 16 rows of 30 kernels each. If $\frac{1}{4}$ of the kernels are removed to make popcorn, how many kernels remain on the cob? [\[Reproduction in Plants\]](#)
- (A) 360
 - (B) 324
 - (C) 400
 - (D) 480
 - (E) 240
 - (F) 120
- 82.** Express the ratio 0.6 : 0.25 in its simplest whole-number form. [\[Comparing Quantities\]](#)
- (A) 24:25
 - (B) 6:25
 - (C) 12:5
 - (D) 3:1
 - (E) 24:10
 - (F) 5:12
 - (G) 12:25
- 83.** About 1 in every 8 forest plant species has a medicinal use. In a forest of 2400 species, how many have a medicinal use, and how many do not? [\[Forests\]](#)
- (A) 300 medicinal, 2100 not
 - (B) 1200 medicinal, 1200 not
 - (C) 150 medicinal, 2250 not
 - (D) 192 medicinal, 2208 not
 - (E) 240 medicinal, 2160 not
 - (F) 2100 medicinal, 300 not
- 84.** The perimeter of a square is equal to the circumference of a circle of radius 7 cm ($\pi = \frac{22}{7}$). Find the side of the square. [\[Perimeter & Area\]](#)
- (A) 33 cm
 - (B) 5.5 cm
 - (C) 7 cm
 - (D) 22 cm
 - (E) 44 cm
 - (F) 11 cm
 - (G) 14 cm

85. Among 200 flowers, 35% self-pollinate, 45% cross-pollinate, and the rest are not pollinated at all. How many flowers are not pollinated? [\[Reproduction in Plants\]](#)

- (A) 90
- (B) 40
- (C) 160
- (D) 70
- (E) 80
- (F) 20
- (G) 36

86. A sum of money doubles itself in 8 years under simple interest. What is the rate of interest per annum?

[\[Comparing Quantities\]](#)

- (A) 16%
- (B) 10%
- (C) 8%
- (D) 100%
- (E) 12.5%
- (F) 11.25%

87. By day a forest makes 900 kg of oxygen per hour, but at night it uses up 200 kg per hour. Over a full day with 10 hours of daylight and 14 hours of night, what is the NET oxygen produced? [\[Forests\]](#)

- (A) 6200 kg
- (B) 2800 kg
- (C) 11800 kg
- (D) 9000 kg
- (E) 5580 kg
- (F) 16800 kg
- (G) 700 kg

88. A wall is 4 m by 3 m and has a door of 2 m by 1 m. Find the area of the wall to be painted (the wall area minus the door). [\[Perimeter & Area\]](#)

- (A) 12 m²
- (B) 9 m²
- (C) 14 m²
- (D) 10 m²
- (E) 11 m²
- (F) 8 m²
- (G) 2 m²

89. A pea pod develops from an ovary that had 8 ovules, but only 6 of them became seeds. What percentage of the ovules became seeds? [\[Reproduction in Plants\]](#)

- (A) 48%
- (B) 62.5%
- (C) 25%
- (D) 75%
- (E) 80%
- (F) 150%
- (G) 60%

90. A fruit-seller buys 100 oranges for Rs 400 and sells them at 5 oranges for Rs 25. Find his profit percent. [\[Comparing Quantities\]](#)

- (A) 25%
- (B) 50%
- (C) 20%
- (D) 15%
- (E) 30%
- (F) 10%

91. A cleared area can graze 5 cows per hectare. If 18 hectares are cleared, how many cows can graze, and how many hectares would be needed for 135 cows? [\[Forests\]](#)

- (A) 90 cows; 675 hectares
- (B) 23 cows; 27 hectares
- (C) 90 cows; 130 hectares
- (D) 13 cows; 27 hectares
- (E) 90 cows; 27 cows
- (F) 100 cows; 27 hectares
- (G) 90 cows; 27 hectares

92. A rectangular garden is 40 m by 25 m and is surrounded by a 2.5 m wide path OUTSIDE it. Find the area of the path. [\[Perimeter & Area\]](#)

- (A) 375 m²
- (B) 325 m²
- (C) 1000 m²
- (D) 350 m²
- (E) 300 m²
- (F) 315 m²

93. A Bryophyllum leaf grows buds along its margin. If 4 leaves each grow 9 buds and $\frac{2}{3}$ of all the buds survive to become plants, how many new plants form? [\[Reproduction in Plants\]](#)

- (A) 24
- (B) 48
- (C) 27
- (D) 18
- (E) 36
- (F) 12
- (G) 13

94. After spending 40% of his money on rent and 25% of the remainder on food, a person is left with Rs 9000. Find the original amount. [\[Comparing Quantities\]](#)

- (A) Rs 25714
- (B) Rs 22000
- (C) Rs 12000
- (D) Rs 36000
- (E) Rs 20000
- (F) Rs 18000

- 95.** A forest basin receives 1200 mm of rain a year. Deforestation cuts the rainfall by $\frac{1}{6}$. What is the new rainfall? [\[Forests\]](#)
- (A) 900 mm
 - (B) 200 mm
 - (C) 1100 mm
 - (D) 1020 mm
 - (E) 1000 mm
 - (F) 720 mm
 - (G) 1400 mm
- 96.** A square field has a side of 84 m. A man walks all the way around it 5 times. How far does he walk, in kilometres? [\[Perimeter & Area\]](#)
- (A) 0.336 km
 - (B) 8.4 km
 - (C) 0.84 km
 - (D) 1.51 km
 - (E) 1680 km
 - (F) 1.68 km
 - (G) 16.8 km
- 97.** A pollen tube grows at 0.5 mm per minute to reach an ovule 30 mm away. How long does it take to reach the ovule? [\[Reproduction in Plants\]](#)
- (A) 30 minutes
 - (B) 6 minutes
 - (C) 60 minutes
 - (D) 90 minutes
 - (E) 54 minutes
 - (F) 150 minutes
 - (G) 15 minutes
- 98.** A's salary is 20% more than B's salary. By what percent is B's salary less than A's? [\[Comparing Quantities\]](#)
- (A) 12%
 - (B) 18%
 - (C) $16\frac{2}{3}\%$
 - (D) 20%
 - (E) 80%
 - (F) 15%
 - (G) 25%
- 99.** Forest soil holds about 4000 seeds per square metre, but after a fire only 12% survive. How many seeds survive per square metre, and how many across 3 square metres? [\[Forests\]](#)
- (A) 48 per m²; 144 in 3 m²
 - (B) 480 per m²; 480 in 3 m²
 - (C) 3520 per m²; 1440 in 3 m²
 - (D) 480 per m²; 1440 in 3 m²
 - (E) 120 per m²; 360 in 3 m²
 - (F) 480 per m²; 14400 in 3 m²
 - (G) 1200 per m²; 3600 in 3 m²

100. A rectangle 16 cm by 9 cm is reshaped into a square of the same area. Find the side and the perimeter of the square. [\[Perimeter & Area\]](#)

- (A) side 144 cm, perimeter 576 cm
- (B) side 12 cm, perimeter 48 cm
- (C) side 25 cm, perimeter 100 cm
- (D) side 12 cm, perimeter 144 cm
- (E) side 12 cm, perimeter 24 cm
- (F) side 12.5 cm, perimeter 50 cm